

Supplementary Information

Interpolation versus generalization in machine learning prediction of micropollutant adsorption on layered double hydroxides

[AUTHORS]

Text S1 Data cleaning rules

All rules are implemented in `ldh_pipeline.load_data` and applied identically in every notebook. (1) Column names are stripped of surrounding whitespace. (2) Categorical labels are normalized by lowercasing, collapsing internal whitespace and hyphen spacing, and stripping trailing punctuation, after which known synonyms are merged (e.g., *co-precipitation or aging*, *co-precipitation* $\rightarrow$ *Co-precipitation*; *reconstruction or memory effect* $\rightarrow$ *Reconstruction*). (3) The removal ratio is converted to percent units by a range-aware rule: if the column maximum is ≤ 1.5 the values are multiplied by 100; an assertion then checks that all values lie in $[0, 100]$, which makes the conversion idempotent under re-execution. (4) Records identical in every descriptor are reduced to their first occurrence (83 records removed). (5) Missing numeric values are imputed inside each model pipeline by a five-neighbor KNN imputer computed in z-score space and mapped back to physical units; columns with no observed value in a training fold fall back to the fold mean. The source-study identifier is retained for grouping only and never enters the feature matrix.

Text S2 Hyperparameter search spaces and selected values

Tree-ensemble hyperparameters were selected by Optuna (TPE sampler, 30 trials, seeded) on cross-validation folds of the training data of the reference split; grid searches used the same folds. Selected values are read directly from the saved model pipelines (`Model/interp_row_42/saved_model`).

Table S1: Hyperparameter search spaces and selected values

Model	Search space	Selected
HistGB	max_iter 100–600; lr 10^{-4} –1 (log); depth 3–15; leaf 3–9	max_iter 484; lr 0.089; c
XGBoost	n_estimators 100–600; lr 10^{-4} –1 (log); depth 3–15; child weight 3–9	495 trees; lr 0.114; dept
RF	n_estimators 100–600; depth 3–15; leaf 3–9	347 trees; depth 13; lea
Cubist	committees 1–9; neighbors none, 1–8	committees 9; neighbors
KNN	k 1–9	$k = 1$
ANN	fixed: 1 hidden layer, 32 ReLU units, Adam, 2000 iterations	—
LR	no hyperparameters	—

Table S2: Leave-one-study-out per-study MAE distribution (RR%), 59 studies, fixed hyperparameters

Model	Median	Q25	Q75
Cubist	14.1	4.7	22.2
KNN	16.0	6.5	30.3
HistGB	16.4	4.7	25.8
RF	16.8	6.1	25.9
ANN	18.1	11.5	38.6
XGB	18.3	3.8	25.4
LR	28.4	18.4	34.8
Global-mean baseline	24.2	—	—
Per-pollutant-mean baseline	25.1	—	—

Text S3 Per-model results under both tasks

Text S4 Extended statistical description

Text S5 Additional figures

Table S3: Wall-clock training and prediction time on the reference split (seconds; single measurement, indicative only)

Model	Fit	Predict
KNN	0.05	0.04
LR	0.14	0.05
Cubist	0.22	0.04
RF	0.37	0.09
ANN	0.67	0.01
HistGB	0.86	0.08
XGBoost	10.1	0.03

Table S4: Extended descriptor statistics

Descriptor	Mean	SD	Median	Range	Skewness	Missing (%)
$\log K_{ow}$	0.83	2.34	-0.02	-2.67-4.51	0.5	0
MW (g/mol)	349.4	97.1	331.3	152.2-733.9	0.5	0
E	2.04	0.75	1.81	0.70-3.60	0.8	0
S	2.33	0.71	2.10	0.70-3.60	0.4	0
A	0.83	0.38	0.70	0.31-1.65	0.8	0
B	1.73	0.94	1.70	0.45-3.50	0.5	0
V	2.41	0.76	2.30	1.13-6.35	2.0	0
TPSA ($\text{\AA}^2$)	101.7	55.1	76.2	37.3-194.0	0.6	0
M(II)/M(III)	3.02	1.21	3.00	0.50-8.00	1.9	0
Avg. atomic weight (g/mol)	42.8	16.2	48.1	24.8-73.7	0.1	0
Avg. d-electrons	3.77	3.69	4.25	0-8.89	0.1	0
Avg. Pauling electronegativity	1.59	0.17	1.59	1.30-1.89	0.4	0
Hydrothermal temp. ($^{\circ}\text{C}$)	78.6	28.6	70	25-160	1.5	13.3
Hydrothermal time (h)	22.9	4.6	24	12-40	-0.1	13.3
Calcination temp. ($^{\circ}\text{C}$)	414	73	400	200-550	-0.3	66.0
Calcination time (h)	4.7	3.0	4	1-12	1.3	66.0
d_{003} (nm)	0.78	0.11	0.77	0.25-1.13	-0.8	2.5
BET (m^2/g)	85.7	57.6	82.0	0.4-225.7	0.5	7.9
PV (cm^3/g)	0.24	0.33	0.10	0-1.46	2.2	25.3
APD (nm)	12.2	12.0	8.4	0.2-64.2	1.6	13.9
pH_{pzc}	8.02	1.56	7.50	5.50-11.56	0.7	24.6
pH	6.50	2.19	6.70	2.0-12.0	0.3	0
Temperature ($^{\circ}\text{C}$)	25.3	2.9	25	10-55	2.6	0
Contact time (h)	9.9	33.3	1.5	0.02-192	4.9	0
Dose (g/L)	2.57	6.38	0.90	0.01-50	6.0	0
C_i (mg/L)	94.4	188.9	40	1-1688	5.0	0

Table S5: Composition of the compilation by pollutant, metal, and synthesis route (records / source studies)

Level	Rec.	St.	Level	Rec.	St.
<i>Micropollutant</i>			<i>Metal(II)</i>		
Diclofenac	304	3	Mg	633	15
Tetracycline	208	8	Zn	312	31
Ciprofloxacin	181	32	Ni	162	7
Doxycycline	106	3	Cu	92	3
Levofloxacin	102	3	ZnCo	58	2
Amoxicillin	94	3	Ca	34	1
Norfloxacin	90	1	NiAl	28	1
Metronidazole	81	1	CoZn	23	1
Methylparaben	56	1	<i>Metal(III)</i>		
Ibuprofen	39	1	Al	819	47
Moxifloxacin	25	1	Fe	311	12
Delafloxacin	23	1	TiAl	153	1
Erythromycin	22	1	Ce	31	1
Oxytetracycline	11	1	Ti	28	1
<i>Synthesis route</i>					
Co-precipitation	1171	54	Urea hydrolysis	132	3
Hydrothermal	28	1	Reconstruction	11	1

Table S6: Variance inflation factors of the numeric descriptors (six largest; 22 of 26 exceed 10)

Descriptor	VIF
Hydrothermal temperature	2.5×10^6
Hydrothermal time	1.6×10^6
d_{003}	1.1×10^6
A	7.8×10^5
Calcination temperature	4.2×10^5
PV	3.4×10^5

Table S7: Share of total mean |SHAP| attribution by descriptor group (HistGB, hold-out test)

Descriptor group	Share (%)
Operating conditions	54.2
Material texture	17.8
Composition and synthesis	14.4
Micropollutant descriptors	13.5

Table S8: Feature-set comparison for XGBoost and Cubist (interpolation mean $\pm$ SD over 20 splits)

Feature set	Model	Interp. R^2	Interp. MAE
Full (30)	XGBoost	0.84 ± 0.03	6.3 ± 0.6
Full (30)	Cubist	0.79 ± 0.03	8.0 ± 0.6
Correlation-screened (21)	XGBoost	0.84 ± 0.03	6.4 ± 0.6
Correlation-screened (21)	Cubist	0.79 ± 0.04	8.2 ± 0.6
Top-10 permutation	XGBoost	0.84 ± 0.03	6.4 ± 0.6
Top-10 permutation	Cubist	0.79 ± 0.03	8.1 ± 0.5

Table S9: Bootstrap 95% confidence intervals for subgroup median RR% differences (5,000 resamples)

Contrast	Median diff.	95% CI	Significant	n_a / n_b
Zn-Al vs Mg-Al	8.7	$[-0.7, 17.2]$	no	128 / 561
Calcined vs uncalcined	10.1	$[1.7, 17.9]$	yes	457 / 885
Diclofenac vs tetracycline	15.5	$[8.3, 27.3]$	yes	304 / 208

Table S10: Screening-threshold classification ($RR\% \geq 80$) from the regression predictions

Task (model)	Precision	Recall	F1	True positives / n
Interpolation (HistGB)	0.92	0.58	0.71	38 / 252
LOSO (Cubist)	0.32	0.06	0.10	203 / 1342
LOSO (HistGB)	0.17	0.01	0.02	203 / 1342

Table S11: Leave-one-study-out median MAE (RR%) by dominant pollutant (≥ 3 studies) and model

Pollutant	ANN	Cubist	HistGB	KNN	LR	RF	XGB
Ciprofloxacin	13.4	5.4	5.0	7.0	28.4	6.4	4.2
Doxycycline	56.0	15.2	16.4	42.4	31.7	26.4	25.5
Amoxicillin	13.2	17.2	28.2	11.9	14.7	29.4	28.4
Tetracycline	43.6	20.7	23.9	24.4	22.3	26.3	23.7
Diclofenac	59.1	21.8	18.3	38.3	71.1	20.0	18.3
Levofloxacin	33.7	22.2	24.0	31.8	30.5	20.8	25.2

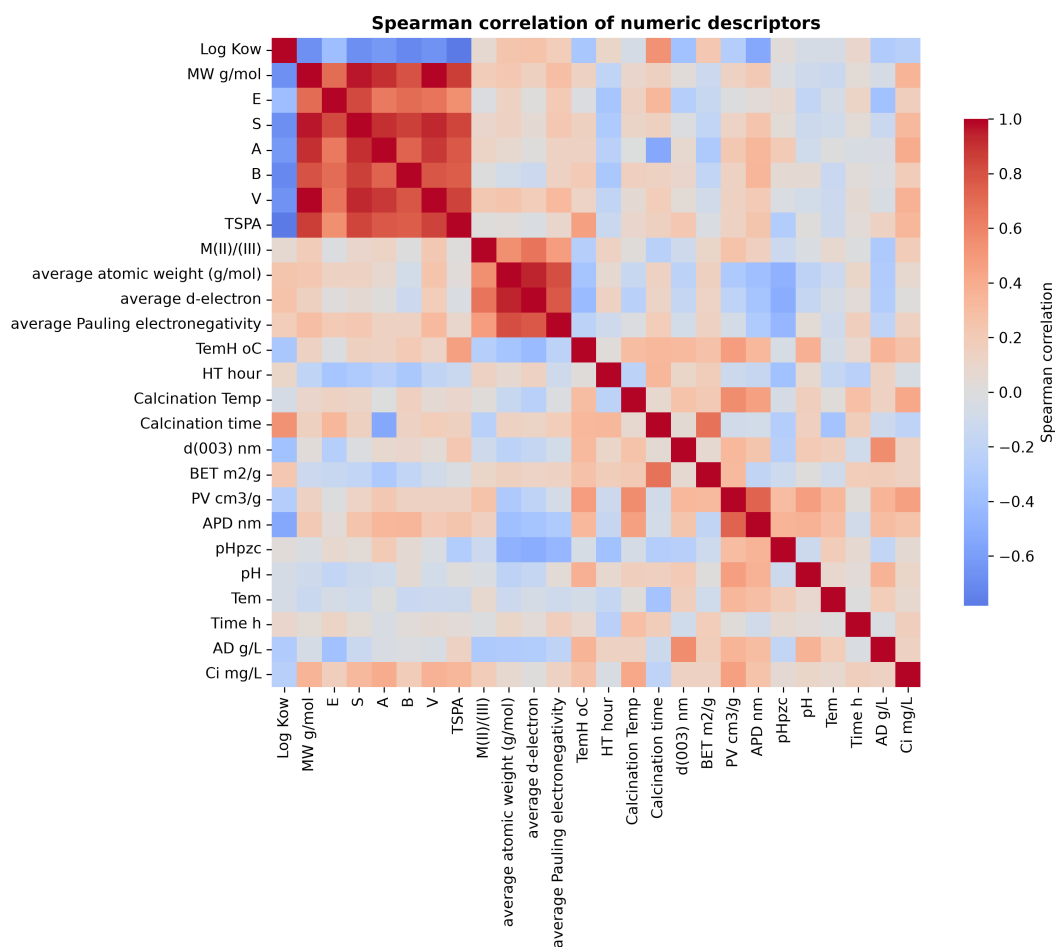


Figure S1: Spearman correlation of the numeric descriptors

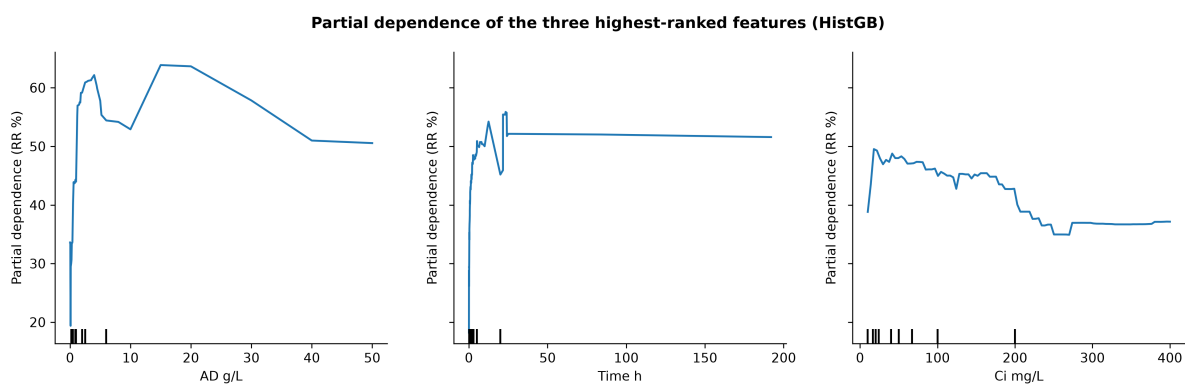


Figure S2: Partial dependence of the three highest-ranked features (HistGB)

SHAP dependence of $\text{pH}-\text{pH}_{\text{pzc}}$, colored by Log Kow (HistGB, test set)

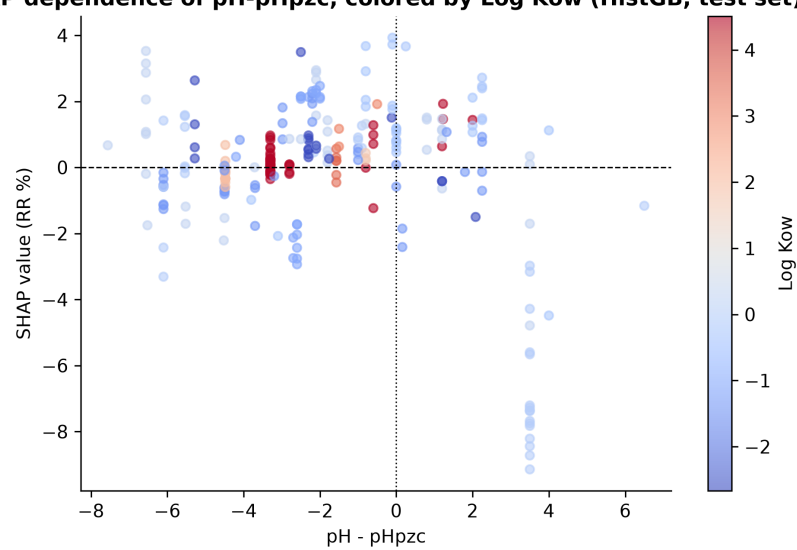


Figure S3: SHAP dependence of $\text{pH}-\text{pH}_{\text{pzc}}$ colored by $\log K_{\text{ow}}$ (HistGB with engineered features, hold-out test set)

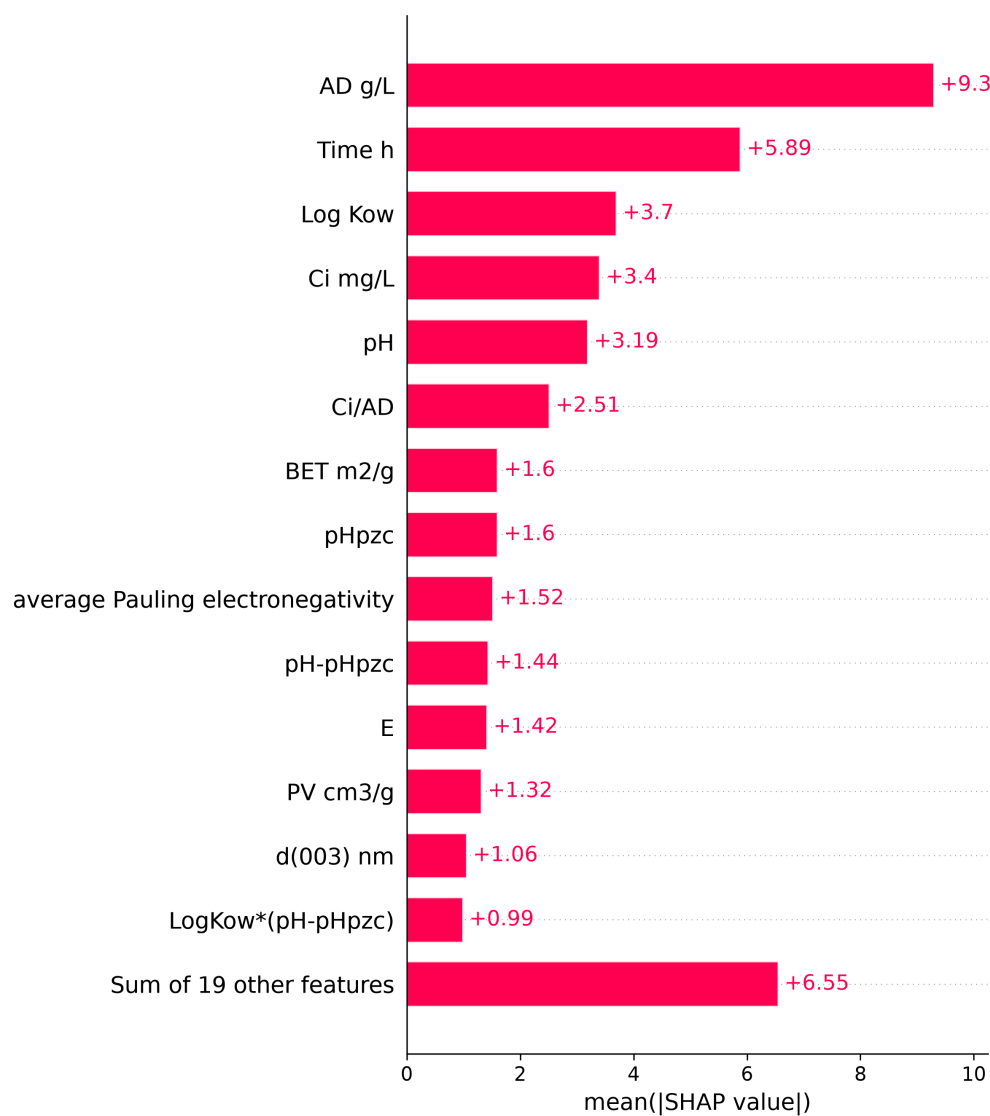


Figure S4: Global SHAP importance, HistGB with engineered features